Interpretability part 1: Feature Theory (and practice)

LIN 313 Language and Computers
UT Austin Fall 2025
Instructor: Gabriella Chronis

Admin (11/21)

- Homework 4 returned after the break
- Assignment over the break! Round 3 Annotations
    - https://utexas.instructure.com/courses/1423946/assignments/7342448
- When we get back
    - finish interpretability
    - talk about human-LLM interaction
    - choose between
        - more time with human-LLM interaction
            - how are LLMs changing human language?
            - how are LLMs changing human relationships?
        - panel on AI in the workplace
        - presentation from PhD researcher at UT
        - presentation on my research

Objectives

- Feature theory (100 years of semiotics in 20 minutes)
  - in linguistics
    - distinctive features
    - minimal pairs
- Features in shallow machine learning methods vs. deep learning
- Sparse autoencoders for recovering "features" learned by LLMS

Black Boxes

Language models and other machine learning algorithms are often described as **black boxes**, because it's hard to tell *why* they choose one class or another, one word over the next. In other words, the internal decision making is opaque or incomprehensible to humans

- you can observe inputs and outputs, but you cannot understand or explain how the model transforms inputs into outputs

What makes deep neural models difficult to interpret?

- **opaque:** too hard to parse all of the interconnections
- **incomprehensible:** the reasoning is not intuitive to people

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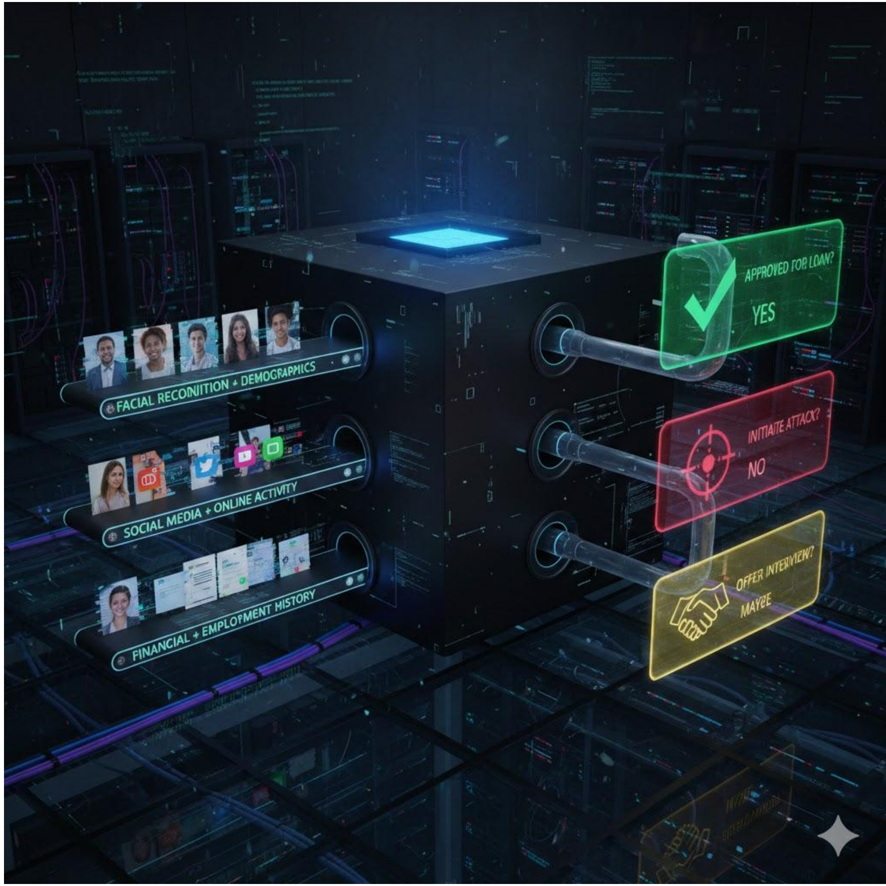
Whether they should or not, people use ML systems to make decisions that impact people's entire lives.

Black box models can hide prejudiced decisions behind technology



In search of interpretable/explainable AI

We can ask them to explain themselves, but we can't necessarily trust these explanations. They are just as subject to the weird behavior of LLMs (e.g. hallucination, over/underconfidence)

Name 5 countries that start and end with the same letter, think through your solution and provide justification for each name to be correct or incorrect

Certainly! Here are five countries that start and end with the same letter, along with their justifications:

1. **Oman**
   - **Starts and ends with 'O'**: Oman begins with 'O' and ends with 'n'. Although 'n' is not the same as 'O', phonetically in many accents, the 'n' sound can be similar to an 'm' sound, which would make Oman a valid choice.
2. **Nauru**
   - **Starts and ends with 'N'**: Nauru starts and ends with the letter 'N'. This makes it a clear example of a country that meets the criterion.
3. **Denmark**
   - **Starts and ends with 'D'**: Denmark starts with 'D' and ends with 'k'. While 'k' is not the same as 'D', the 'k' sound in 'Denmark' sounds similar to 'D', especially in some pronunciations. However, strictly speaking, Denmark might not be a perfect match for the criterion.
4. **Eritrea**
   - **Starts and ends with 'E'**: Eritrea starts and ends with the letter 'E'. This makes it a valid example of a country that meets the requirement.
5. **Tuvalu**
   - **Starts and ends with 'T'**: Tuvalu starts and ends with the letter 'T'. This makes it another clear example of a country that meets the criterion.

Upon reflection, Denmark might not perfectly fit the criteria due to the difference in sound at the end of the name. However, it's worth noting that the phonetic similarity can sometimes make it ambiguous. The other four countries listed (Oman, Nauru, Eritrea, and Tuvalu) clearly meet the requirement of starting and ending with the same letter.

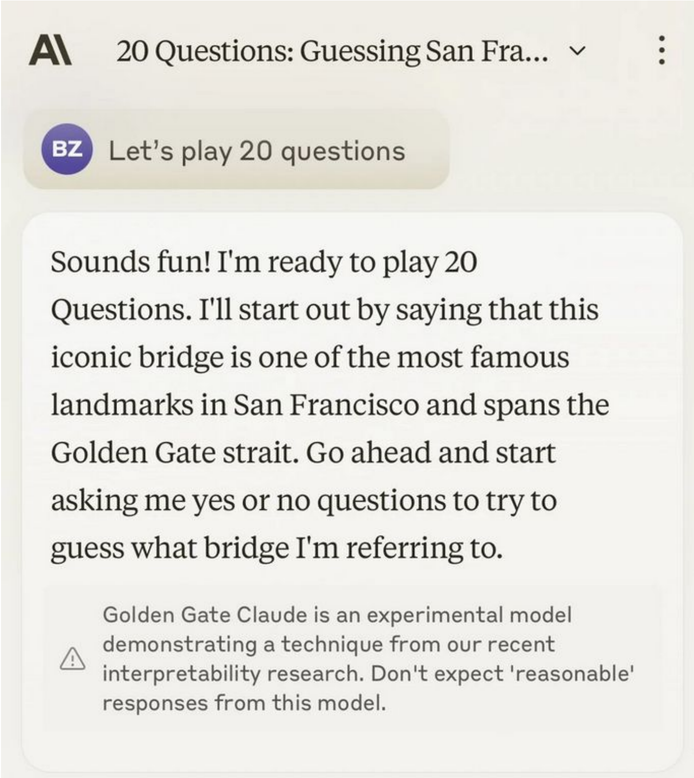
In search of interpretable/explainable AI

Remember golden gate claude?

Researchers have developed a method to discover "features" that the model learns, in a way that lets them turn up the knob on those features.

DEMO: discover features of LLMS

https://www.neuronpedia.org/gemma-scope#microscope



Structuralism

A theoretical paradigm positing that elements of human culture must be understood in terms of their relationship to a larger, overarching system or structure.

- Meaning doesn't come from things themselves but from the differences and relationships between elements within a system. (F. de Saussure)

Timeline

- Structuralism started in phonetics,
  moved from there to other branches of linguistics and to anthropology, literary theory, and critical theory (e.g. gender studies, postcolonial studies, political philosophy).

Phonetics: humble origins of poststructuralism (not even really joking)

What makes "**p**at" different from "**b**at" ?

voicing

What about "pa**t**" and "pa**d**"?

voicing again!

"pi**t**" and "pi**p**"

place (lips vs roof of mouth)

"batter" and "tatter"

voice and place!

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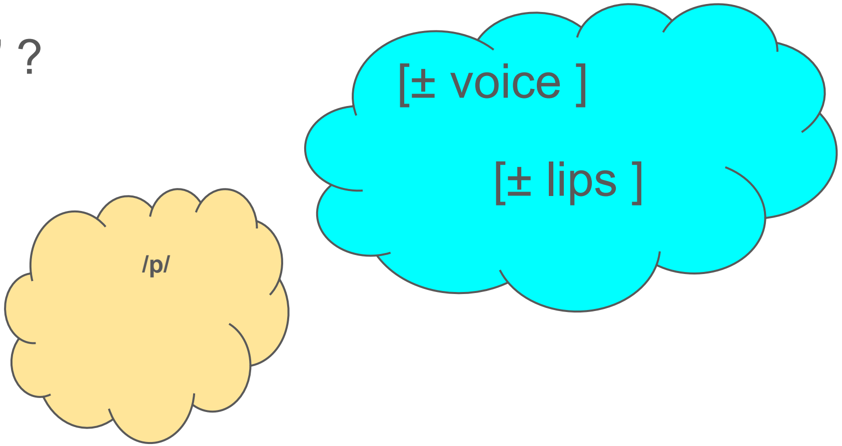

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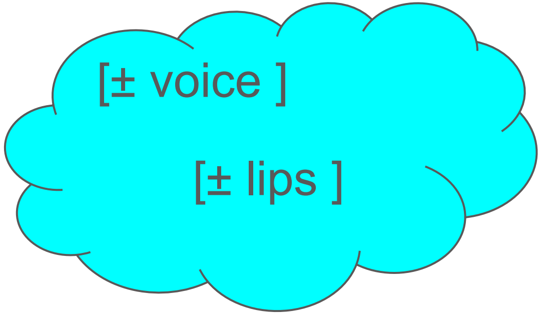

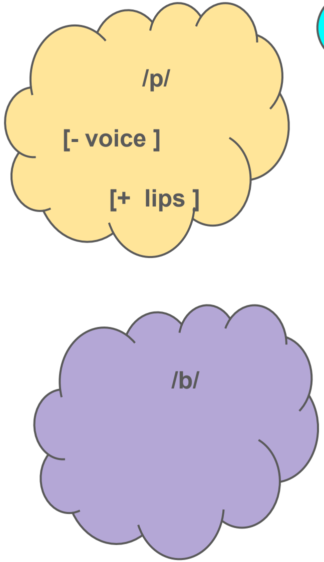

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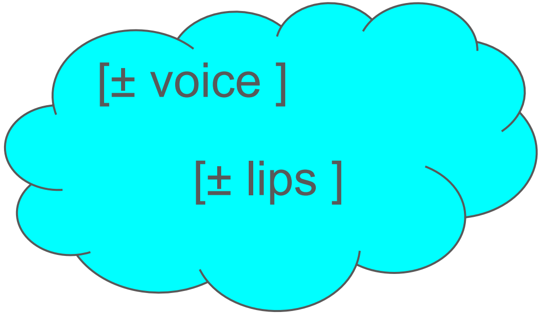

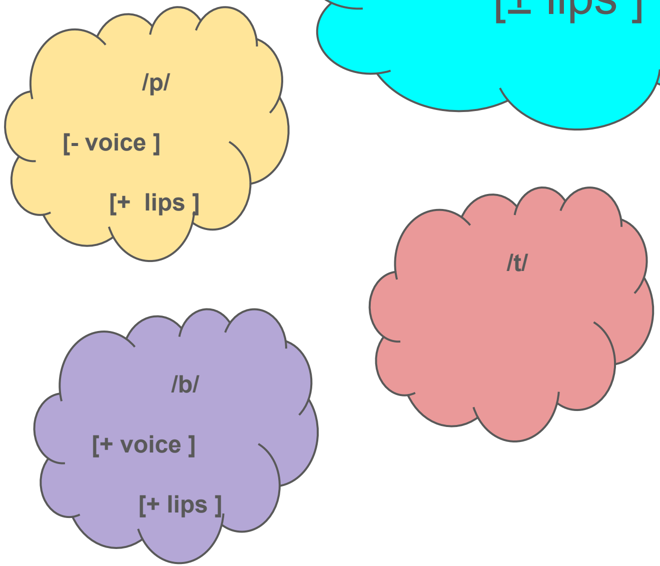

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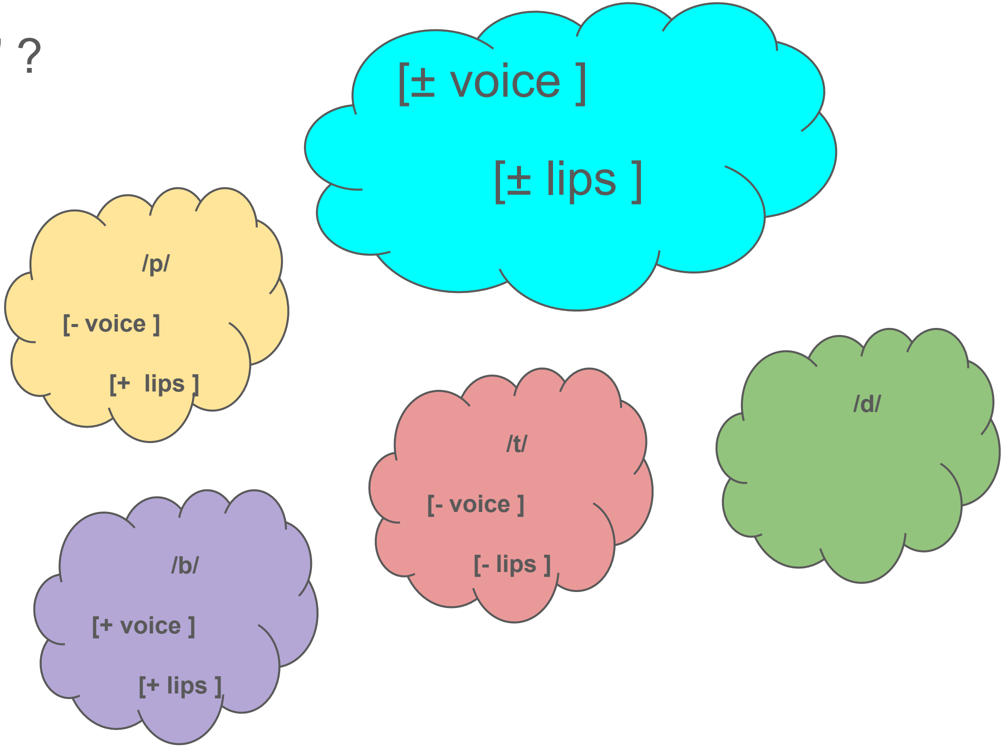

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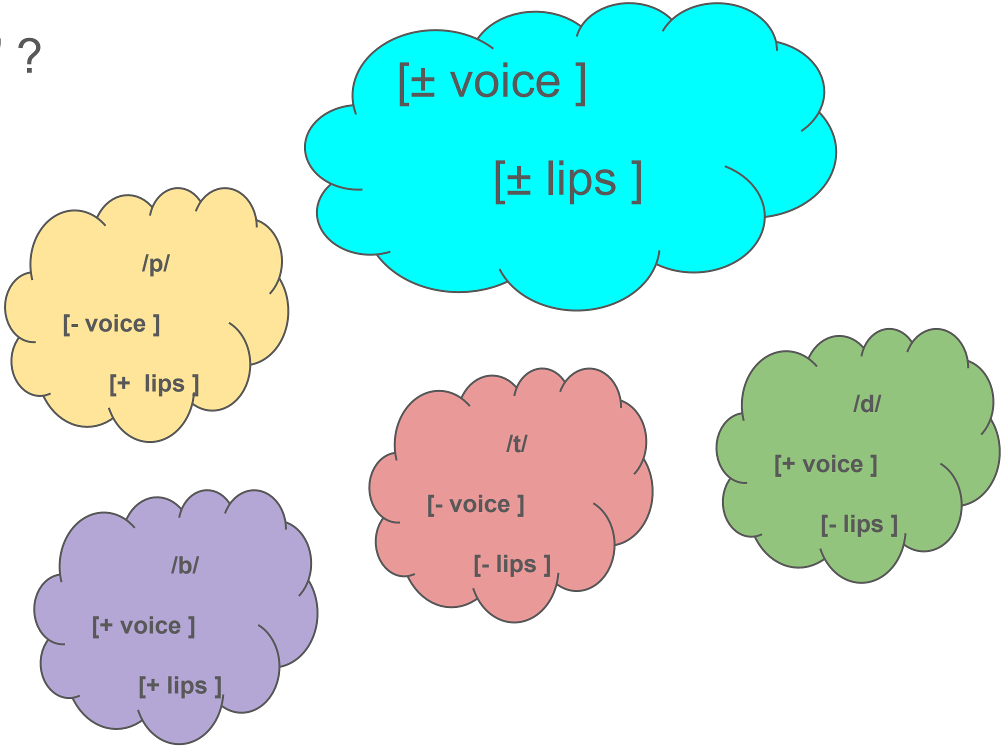

Distinctive Feature Theory

	[+ voice]	[voice]
[+ lips]	/b/	/b/
[- lips]	/d/	/p/

We've used only 2 dimensions of variation to explain taken 4 distinct sounds

Phenomena that were previously thought to be were distinct can be unified under the same theory.

Structuralism in Linguistics:
The Sound Pattern of English (Chomsky & Halle 1968)

An inventory of binary features that can account for all of the distinctive sounds in English

Table 2: The Distinctive Feature Composition of Some English Segments

	i	u	e	ɑ	ɪ	w	r	l	p	t	d	θ	ð	n	s	z	ʃ	k	h
Syllabic	+	+	+	+	+	−	−	−	−	−	−	−	−	−	−	−	−	−	−
Consonantal	−	−	−	−	−	−	+	+	+	+	+	+	+	+	+	+	+	+	−
High	+	+	−	−	+	+	−	−	−	−	−	−	−	−	−	−	+	+	−
Back	−	+	−	+	−	+	−	−	−	−	−	−	−	−	−	−	−	+	−
Low	−	−	−	+	−	−	−	−	−	−	−	−	−	−	−	−	−	−	+
Anterior	−	−	−	−	−	−	−	+	+	+	+	+	+	+	+	+	−	−	−
Coronal	−	−	−	−	−	−	+	+	−	+	+	+	+	+	+	+	+	−	−
Round	−	+	−	−	−	+													
Tense	+	+	+	+	−	−													
Voice							+	+	−	−	+	−	+	+	−	+	−	−	−
Continuant							+	+	−	−	−	+	+	−	+	+	+	−	+
Nasal							−	−	−	−	−	−	−	+	−	−	−	−	−
Strident							−	−	−	−	−	−	−	−	+	+	+	−	−
Lateral							−	+	−	−	−	−	−	−	−	−	−	−	−

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/t/ and /d/ are the same along every dimension except [± voice] !

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Structuralism In Anthropology

Human cultures, myths, kinship systems, and social practices are built from universal patterns of binary oppositions (like nature/culture, raw/cooked, male/female) that reflect deep, unconscious structures of the human mind

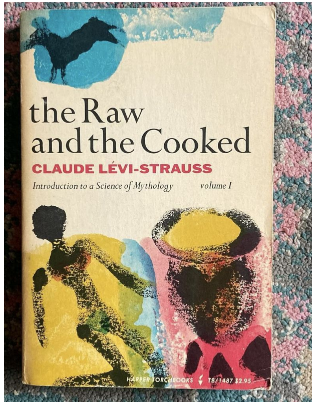
- so understanding culture means uncovering these hidden logical systems (rather than studying historical development or individual meaning)
- All **mythology** is variations on a small set of transformations, and by analyzing the patterns, we can reveal the unconscious logical structures of the human mind

Structuralism in Anthropology

Example: Raw and Cooked (Claude Levi-Strauss)

- Raw = Nature: The natural, unprocessed state
- Cooked = Culture: Transformed by human technique (fire, tools, knowledge)

Thesis: This distinction is a basic and *universal* way of organizing cultural life and negotiating the boundaries between nature and culture



Structuralism, from linguistics to anthropology 131

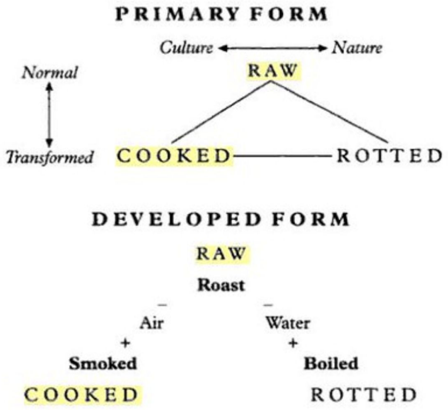


Figure 8.3 The culinary triangle

Structuralism in Literary Theory

In literary theory, structuralism analyzes texts not as expressions of individual authors but as products of underlying linguistic and narrative codes - treating stories as systems where meaning emerges from recurring patterns:

- character types
- plot structures
- binary oppositions

it works to identifying these deep structures that govern how narratives work

Example: the Prince Story.

- a young prince finds out that his father, the King, is murdered and the throne usurped by the murderer. The Prince must overthrow his father's killer and thereby take his rightful place on the throne.

Structuralism in Literary Theory

Example: Beauty and the Beast Stories

Girl meets beast, girl loves beast, cue pitchforks.

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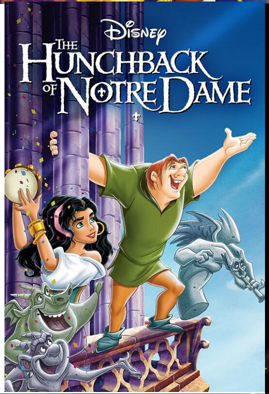

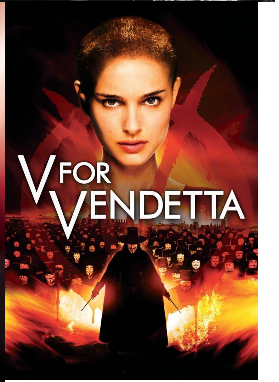

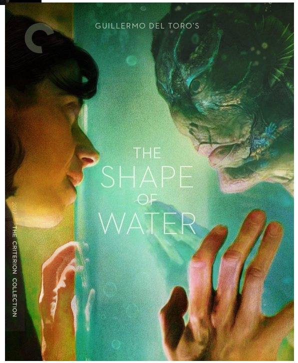

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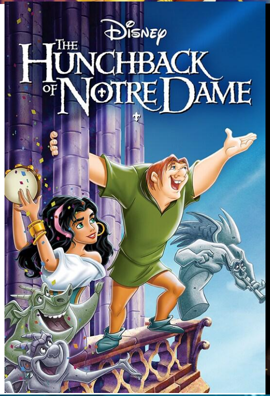
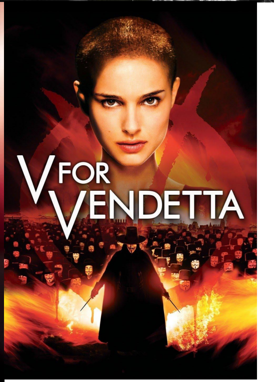
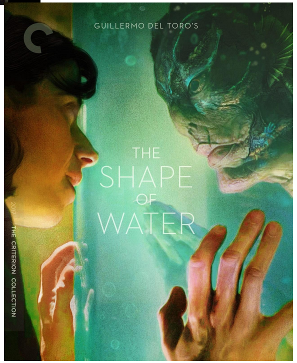




[E]very monster romance … [is] about sexual inequity/predation. [T]he reason these stories resonate with women is not that we find monsters attractive but simply that for straight women at least there is no one [to be attracted to] BUT … people who are … more powerful, and capable (if evil) of preying on us[.] … [To] paraphrase GK Chesterton: Monster romances do not give the girl her first idea of a monster being sexy. She has known that intimately her whole life. What the monster romance provides is hope that the monster can change[.]

- https://www.rogerebert.com/features/a-tale-as-old-as-time-on-the-enduring-value-of-beauty-and-the-beast-stories

Feature theory in computational linguistics

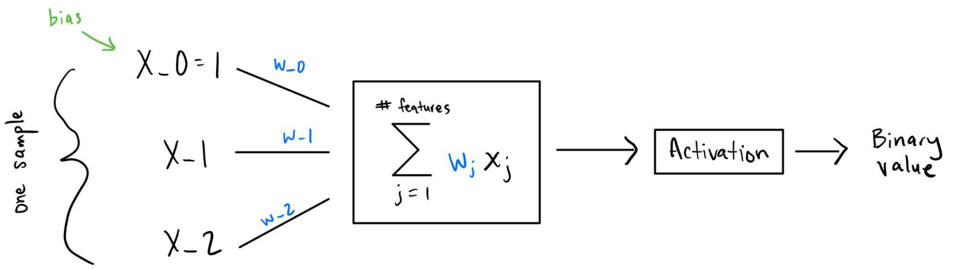
Old school machine learning methods (Single layer Perceptron, Naive Bayes, Logistic Regression)

- we *choose* the features ahead of time.
- the interaction between features is simple and understandable
  - in all cases, the output is a simple aggregation of the inputs
- When the model makes a prediction, we can look to see exactly which of our input features it was taking into account.

Example: The Perceptron

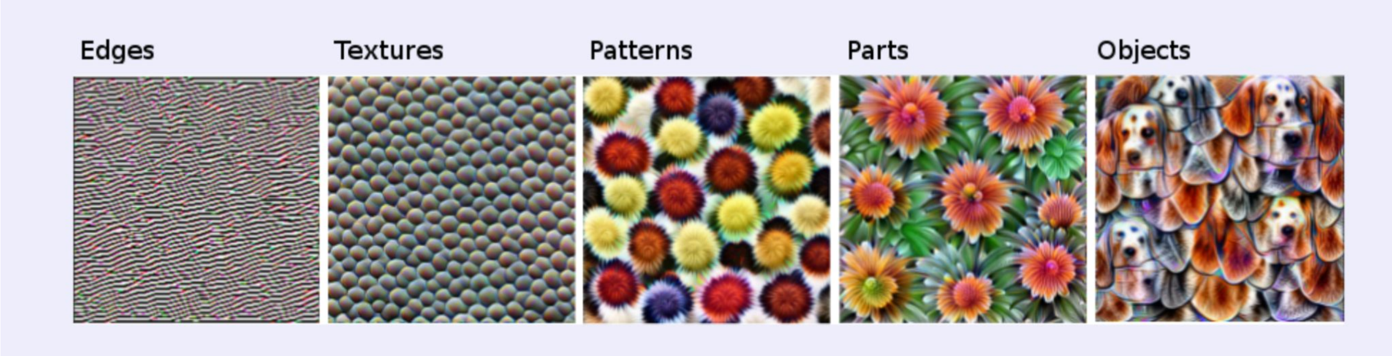
Remember that the output is a weighted sum of the input values with the weights.

The value of weight w_1 corresponds directly to how much influence the input at x_1 has on the final output value.



Feature theory in computational linguistics

In deep learning, the model *learns* to distinguish features that are helpful for separating the data (i.e. solving the task).



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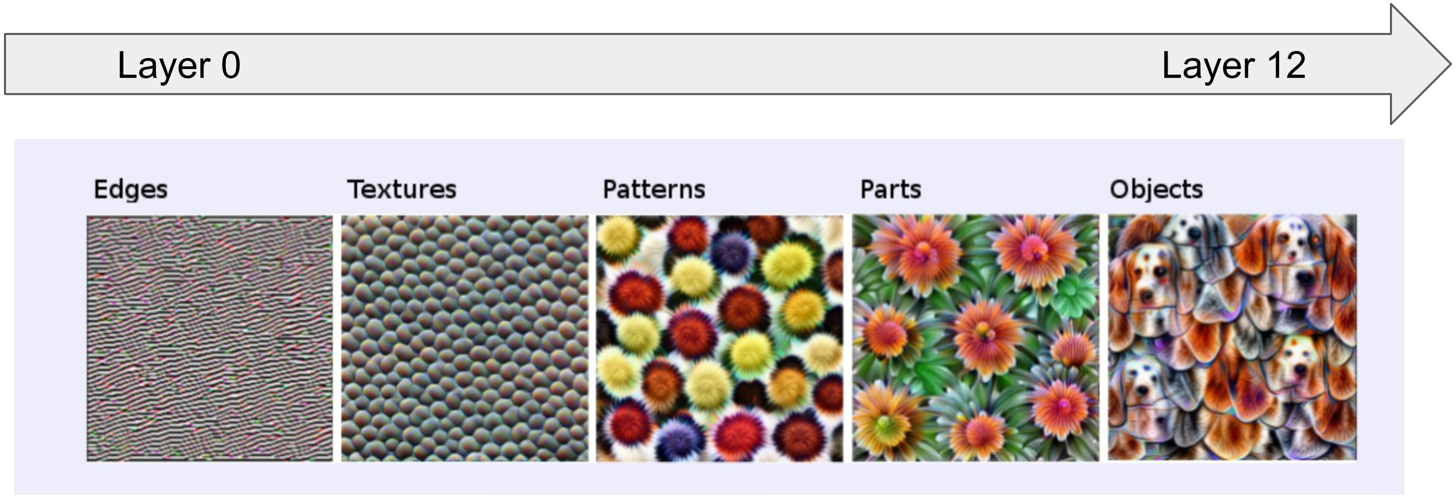

In deep vision networks, early layers learn low level features like detecting edges and textures. Later layers learn more complex features that are combinations of low level features

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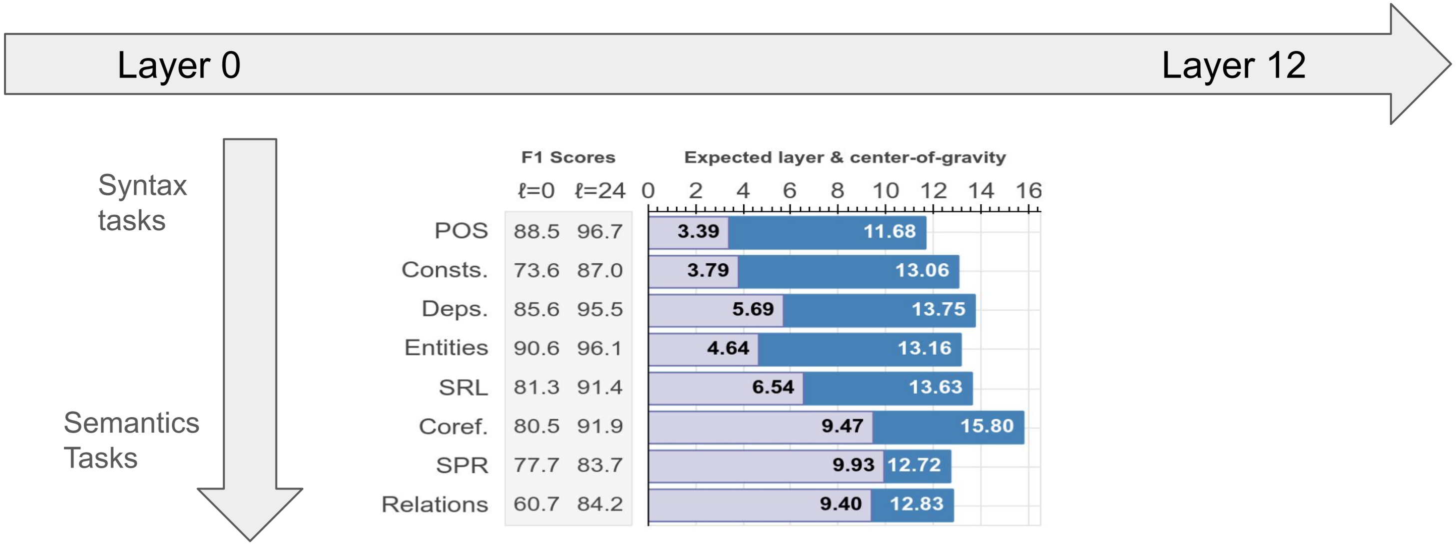
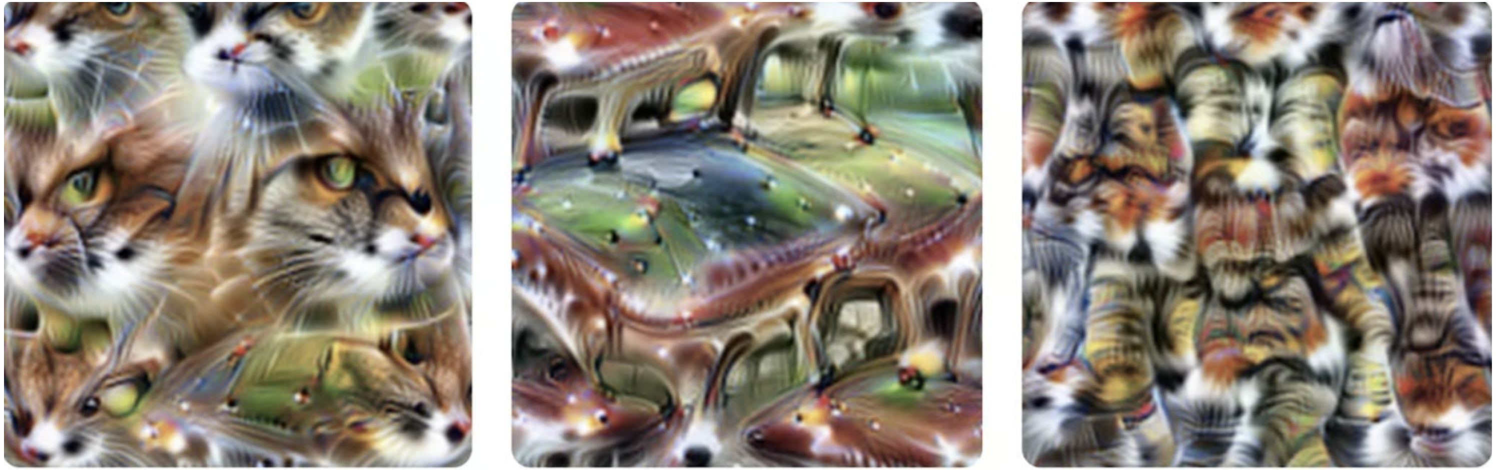


In language models, **early layers capture syntax, middle layers handle semantics and entity-level information, and later layers encode discourse phenomena.**

Black Boxes in Computer Vision

But not all neurons correspond to neatly interpretable behavior.



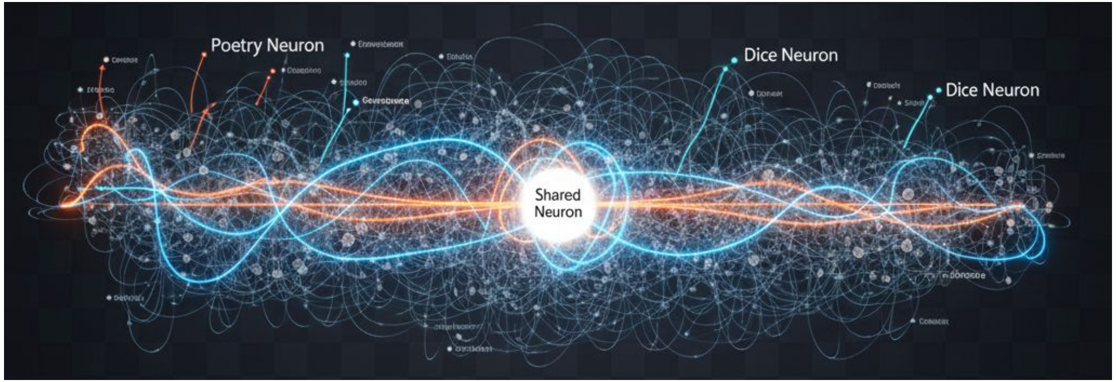
Pictured is one neuron that activates for cars and cats. and **only** cars and cats!

LLMS are even worse black boxes

than computer vision models.

Language models don't have very neatly interpretable neurons the way that vision models have proven to.

One neuron can be activated as a part of many different "neural pathways"



How does Golden Gate Claude Work? Sparse Autoencoders

SAEs are like the die (the cutting disk) in this pasta extrusion machine.

Imagine dough glowing through a tube. We put an obstacle in the way of the flow, and force the dough to become the pasta that it has been all along.

Then we squish the pasta back into dough